Project Title:

Customer Feedback Sentiment Analysis using Azure AI and Python

Name - MANISH KUMAR

Profession - Data Scientist

Problem Statement

Businesses receive large volumes of customer feedback daily, making it difficult to manually analyze sentiments and identify key issues. There is a need for an automated system to process customer reviews and extract meaningful insights.

Objective

- Analyze customer feedback using AI.
- Classify sentiment into **Positive, Negative, Neutral**.
- Process multiple inputs (batch processing).
- Generate actionable business insights.

Tools & Technologies

- Python
- Azure AI Language (Text Analytics)
- Jupyter Notebook

Approach

Step 1: Data Input

Customer feedback text is provided as input.

Step 2: Azure Integration

Azure Text Analytics API is used to process text data.

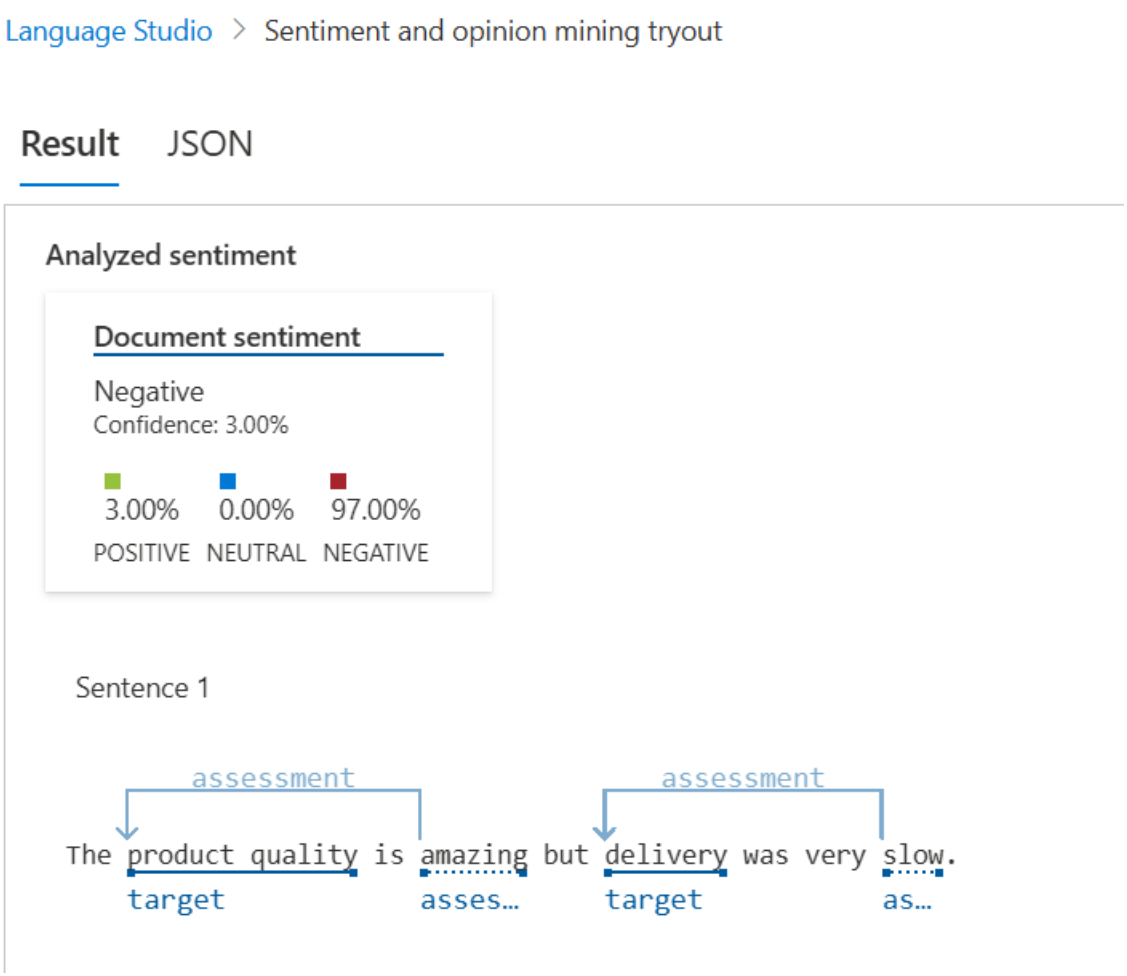
Step 3: Sentiment Analysis

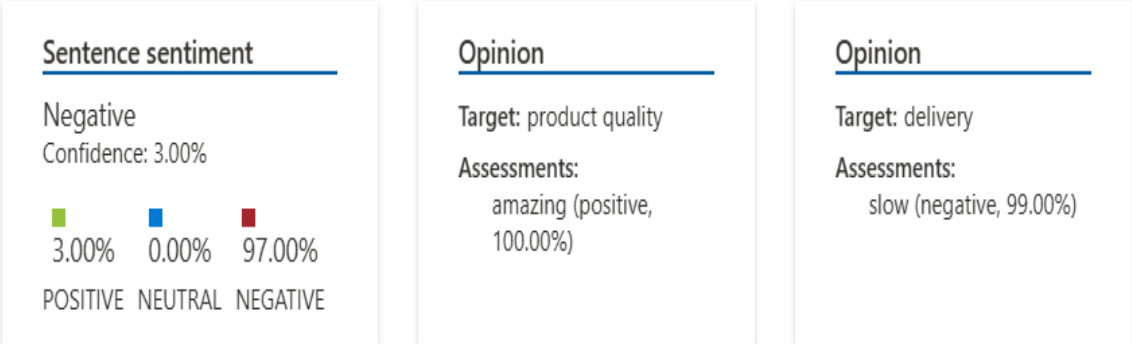
Each feedback is analyzed to determine:

- Overall sentiment
- Confidence scores (positive, neutral, negative)

Step 4: Batch Processing

Multiple feedbacks are processed together for scalability.





Sentiment Analysis using Azure Language Studio

Implementation

Code Snippet 1:

```
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential

# 🔑 Your Azure credentials
key = "Cj8diRQaZThSIHdpjO4i2QQiURuuaOSNyqztqNnK5BeBeq1HXpPsJQQJ99CDACGhslBXJ3w3AAAaACOGCJQv"
endpoint = "https://manish-ds-work.cognitiveservices.azure.com/"

# Create client
client = TextAnalyticsClient(endpoint=endpoint, credential=AzureKeyCredential(key))

# Input text
documents = ["The product quality is amazing but delivery was very slow."]

# Analyze sentiment
response = client.analyze_sentiment(documents)

# Print results
for doc in response:
    print("Overall Sentiment:", doc.sentiment)
    print("Positive Score:", doc.confidence_scores.positive)
    print("Neutral Score:", doc.confidence_scores.neutral)
    print("Negative Score:", doc.confidence_scores.negative)
```

Code Snippet 2:

```python
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential

# 🔑 Your Azure credentials
key = "EeN7RzBLgc7PnV2Bpyw3eCa3fsH4QLPN7dCvqoZtAp1T3mLVeuQNJQQJ99CDACGhslBXJ3w3AAAaACOGHJxg"
endpoint = "https://manish-ds-work.cognitiveservices.azure.com/"

# Create client
client = TextAnalyticsClient(endpoint=endpoint, credential=AzureKeyCredential(key))

# Multiple texts (batch)
documents = [
    "The product quality is amazing but delivery was very slow.",
    "Customer support was very helpful and resolved my issue quickly.",
    "The app crashes again and again. Very frustrating experience.",
    "Average service, nothing special."
]

# Analyze sentiment
response = client.analyze_sentiment(documents)

# Print results
for i, doc in enumerate(response, start=1):
    print(f"\nText {i}:")
    print("Sentiment:", doc.sentiment)
    print("Positive Score:", round(doc.confidence_scores.positive, 2))
    print("Neutral Score:", round(doc.confidence_scores.neutral, 2))
    print("Negative Score:", round(doc.confidence_scores.negative, 2))
```

Python Code for Azure Sentiment Analysis

Output

```
Overall Sentiment: negative
Positive Score: 0.03
Neutral Score: 0.0
Negative Score: 0.97
```

```
Text 1:
Sentiment: negative
Positive Score: 0.03
Neutral Score: 0.0
Negative Score: 0.97

Text 2:
Sentiment: positive
Positive Score: 1.0
Neutral Score: 0.0
Negative Score: 0.0

Text 3:
Sentiment: negative
Positive Score: 0.0
Neutral Score: 0.0
Negative Score: 1.0

Text 4:
Sentiment: neutral
Positive Score: 0.0
Neutral Score: 1.0
Negative Score: 0.0
```

Sentiment Analysis Output from Python

Key Insights

- Customers are satisfied with **product quality and support**
- Major issues found in:
  - Delivery delays
  - Application performance

Business Impact

- Helps companies quickly identify customer pain points
- Reduces manual effort in feedback analysis
- Enables data-driven decision making

Future Enhancements

- Build a **dashboard (Power BI / Streamlit)**
- Add **real-time API integration**
- Extend to **chatbot using CLU / LLM**